Residues modulo 16 for a series defined by a bilateral theta-type equation

A proof of three conjectures in OEIS A355872

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Abstract

The sequence A355872 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, lists the coefficients $a(n)$ of the power series $A(x) = \sum_{n\geq 1} a(n)x^{4n-3} = 2x + 14x^5 + 434x^9 + \cdots$ determined by the equation $x = \sum_{j\in\mathbb{Z}} (-x)^{j^2} A(x)^{(j-1)^2}$, a sum over all integers j. The entry conjectures that every $a(n)$ is 2 modulo 4, and that $a(n)$ is 2 or 6 modulo 8 according as n is odd or even. We prove the stronger statement $a(n) \equiv 2(-1)^{n-1} \pmod{16}$ for every $n \geq 1$. The proof first shows that the equation has exactly one solution over every commutative ring and that only the degrees $4n-3$ occur. Modulo 16 the series $2x/(1+x^4)$ satisfies the equation, because its fourth power vanishes and only three terms of the infinite sum survive. The results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [2] is a public database of integer sequences. Besides the terms, many entries record formulas and *conjectures*: patterns observed in the data that the contributor did not prove. This paper concerns entry A355872 [1], contributed by Paul D. Hanna in 2022, which reads in part:

```
G.f.  A(x) satisfies:  x = Sum_{n=-oo..+oo} (-x)^(n^2) * A(x)^((n-1)^2).

Conjecture:  a(n) == 2 (mod 4) for n >= 1.

Conjecture:  a(2*n-1) == 2 (mod 8) for n >= 1.
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Conjecture:  a(2*n) == 6 (mod 8) for n >= 1.

G.f.  A(x) = Sum_{n>=1} a(n) * x^(4*n-3) satisfies:  ...
```

— OEIS A355872, Paul D. Hanna, August 2022

Here "G.f." stands for *generating function*, the power series whose coefficients are the terms of the sequence. Written with j for the summation index, the defining equation is

$$x = \sum_{j\in\mathbb{Z}} (-x)^{j^2} A(x)^{(j-1)^2}, \qquad A(x) = \sum_{n\geq 1} a(n)\, x^{4n-3}. \tag{1}$$

The sum runs over *all* integers j, positive and negative; such a sum is called *bilateral*. It has the shape of the classical theta series $\sum_{j\in\mathbb{Z}} q^{j^2}$, which is why equations of this kind are called theta-type. The first terms of the sequence are 2, 14, 434, 17662, 829314, 42293582, … (from $n = 1$). Section 2 explains why the infinite sum in (1) makes sense, and Theorem 3.1(a) shows that it determines A.

1.2 The question

The three conjectures concern the remainders of $a(n)$ on division by 4 and by 8. The table shows the remainders modulo 8 and modulo 16 for the first 32 terms.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$a(n)$ mod 8	2	6	2	6	2	6	2	6	2	6	2	6	2	6	2	6
$a(n)$ mod 16	2	14	2	14	2	14	2	14	2	14	2	14	2	14	2	14

n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
$a(n)$ mod 8	2	6	2	6	2	6	2	6	2	6	2	6	2	6	2	6
$a(n)$ mod 16	2	14	2	14	2	14	2	14	2	14	2	14	2	14	2	14

Modulo 16 the remainders alternate between 2 and 14, and this is what we prove.

1.3 What this paper proves

Main result (Theorem 3.1). *Equation* (1) *has exactly one solution, only the degrees* $4n-3$ *occur in it, and for every* $n \geq 1$

$$a(n) \equiv 2\,(-1)^{n-1} \pmod{16}.$$

Reducing modulo 8 and modulo 4 gives all three conjectures of the entry (Corollary 3.2). The key step is the identity

$$A(x) \equiv \frac{2x}{1+x^4} = 2x - 2x^5 + 2x^9 - \cdots \pmod{16}.$$

1.4 What was known

The entry records an asymptotic formula for $a(n)$ (due to Vaclav Kotesovec) and two product formulas obtained from the Jacobi triple product identity. When the entry was checked on 24 October 2026, the three congruences were still labelled as conjectures. We are not aware of an earlier proof.

1.5 Motivation

Theta-type sums such as $\sum_{j\in\mathbb{Z}} q^{j^2} z^j$ are classical objects of combinatorics and number theory; through the Jacobi triple product identity they are connected with partitions and with sums of squares. Here such a sum is used to define a power series implicitly, and the coefficients that result grow quickly (roughly like 70^n). The congruences show that, modulo 16, the coefficients reduce to a two-term periodic pattern. The proof illustrates a mechanism that works in other settings: modulo a power of 2, a series all of whose coefficients are even has a power equal to zero, so an infinite sum can reduce to finitely many terms.

1.6 How to read this paper

Section 2 recalls power series, congruences and bilateral sums. Section 3 states the results and Section 4 proves them. Section 5 describes the computer checks and the formal proof, and Section 6 discusses the scope of the result.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [4, 3]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, is called *normalized*.

2.2 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.3 Sums over all integers

For a series F with coefficients in a commutative ring R and an integer j, put

$$T_j(F) = (-x)^{j^2} F^{(j-1)^2}.$$

Both exponents are nonnegative integers, so $T_j(F)$ is a power series. Because of the factor $(-x)^{j^2}$, its coefficients of degree less than j^2 are zero. Hence a given coefficient of degree d receives contributions only from the finitely many j with $j^2 \leq d$, and the bilateral sum

$$\Theta(F) = \sum_{j \in \mathbb{Z}} T_j(F)$$

is well defined coefficient by coefficient. Equation (1) says that $\Theta(A) = x$. No question of convergence arises: every coefficient of $\Theta(F)$ is a finite sum.

The entry also gives the equation with $(j+1)^2$ in place of $(j-1)^2$; replacing j by $-j$ shows that the two forms are the same.

3 Results

Theorem 3.1. *(a) Over every commutative ring R there is exactly one power series A_R with $\Theta(A_R) = x$.*

(b) The integer solution $A = A_{\mathbb{Z}}$ has nonzero coefficients only in degrees $4n-3$, $n \geq 1$; so $A(x) = \sum_{n \geq 1} a(n) x^{4n-3}$ as in (1).

(c) Modulo 16,

$$A(x) \equiv \frac{2x}{1+x^4} \pmod{16}, \qquad \textit{equivalently} \qquad a(n) \equiv 2\,(-1)^{n-1} \pmod{16} \quad (n \geq 1).$$

Corollary 3.2 (Hanna's conjectures)**.** *For every $n \geq 1$: $a(n) \equiv 2 \pmod 4$, $a(2n-1) \equiv 2 \pmod 8$, and $a(2n) \equiv 6 \pmod 8$.*

Proof. By Theorem 3.1(c), $a(n)$ leaves remainder 2 on division by 16 when n is odd and remainder 14 when n is even. Reducing, the remainder modulo 8 is 2 for odd n and 6 for even n, and modulo 4 it is 2 in both cases. □

4 Proofs

The idea in brief. The term with $j = 0$ in the bilateral sum is A itself. Moving all other terms to the other side expresses A through terms that each carry at least one extra factor of x, so the equation determines the coefficients one at a time, over every ring. Modulo 16, the series $P = 2x/(1 + x^4)$ has all coefficients even, so $P^4 = 0$; in $\Theta(P)$ only the terms with $(j-1)^2 < 4$, that is $j = 0, 1, 2$, survive, and they add up to x.

Throughout, R is a commutative ring and $\Phi(F) = x - \big(\Theta(F) - F\big)$, so that $\Theta(F) = x$ is equivalent to $F = \Phi(F)$.

Lemma 4.1. *Let $r \geq 0$ and $F, G \in R[[x]]$. If $F - G = O(x^r)$, then $\Phi(F) - \Phi(G) = O(x^{r+1})$.*

Proof. For every $e \geq 0$, $F^e - G^e = (F-G)(F^{e-1} + F^{e-2}G + \cdots + G^{e-1})$ is a multiple of $F - G$, hence $O(x^r)$. For $j \neq 0$ we have $j^2 \geq 1$, so $T_j(F) - T_j(G) = (-x)^{j^2}\big(F^{(j-1)^2} - G^{(j-1)^2}\big) = O(x^{r+1})$. The term $j = 0$, which equals $F - G$, does not occur in $\Phi(F) - \Phi(G)$. Each coefficient of $\Phi(F) - \Phi(G)$ is a finite sum of coefficients of the differences $T_j(F) - T_j(G)$ with $j \neq 0$, which proves the claim. □

Proof of Theorem 3.1(a). By Lemma 4.1, the coefficient of x^d in $\Phi(F)$ depends only on the coefficients of F of degree below d. Choosing the coefficients of A_R one at a time so that $A_R = \Phi(A_R)$ holds in each degree therefore gives a solution, and two solutions must agree in every degree. Equivalently, starting from $F_0 = 0$ and putting $F_{k+1} = \Phi(F_k)$, Lemma 4.1 gives $A_R - F_k = O(x^k)$, so each coefficient of F_k is constant from some k on and equals the corresponding coefficient of A_R. □

No division was used, so reducing the integer solution modulo m gives a solution over $\mathbb{Z}/m\mathbb{Z}$, and by uniqueness it is the solution there (Section 2).

Proof of Theorem 3.1(b). Say that a series is *supported in class* 1 if its only nonzero coefficients are in degrees $\equiv 1 \pmod 4$. If F is supported in class 1, then F^e is supported in degrees $\equiv e \pmod 4$, and $T_j(F)$ in degrees $\equiv j^2 + (j-1)^2 \pmod 4$. Of the two consecutive integers $j - 1$ and j, one is even and one is odd, so $j^2 + (j-1)^2 \equiv 0 + 1 = 1 \pmod 4$. Hence $\Theta(F)$, and therefore $\Phi(F)$, is again supported in class 1. All the approximations F_k from the proof of part (a) are thus supported in class 1, and so is their limit A. This also shows that A has constant term 0. □

Proof of Theorem 3.1(c). Work over $R = \mathbb{Z}/16\mathbb{Z}$ and put

$$P(x) = \frac{2x}{1+x^4} = 2\sum_{k\geq 0}(-1)^k x^{4k+1}.$$

Every coefficient of P is even, so every coefficient of P^4 is divisible by $2^4 = 16$; that is, $P^4 = 0$ in $R[[x]]$, and then $P^e = 0$ for all $e \geq 4$. Now $(j-1)^2 \geq 4$ unless $j \in \{0, 1, 2\}$, so all terms of $\Theta(P)$ vanish except

$$T_0(P) = P, \qquad T_1(P) = -x, \qquad T_2(P) = (-x)^4 P = x^4 P.$$

Hence

$$\Theta(P) = P - x + x^4 P = (1 + x^4)P - x = 2x - x = x.$$

By part (a), P is the unique solution over $\mathbb{Z}/16\mathbb{Z}$, so it is the reduction of A modulo 16. The coefficient of $x^{4n-3} = x^{4(n-1)+1}$ in P is $2(-1)^{n-1}$, which gives the formula for $a(n)$. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed the solution of (1) by the fixed-point iteration of the proof, exactly up to degree 120 (that is, $n \leq 30$) and modulo 16 up to degree 1200. The computation reproduces all 17 terms printed in the OEIS entry, confirms that only the degrees $4n-3$ occur, and confirms $a(n) \equiv 2(-1)^{n-1} \pmod{16}$ for $n \leq 300$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [7]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [5] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [6], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A355872.lean` of the repository, in the namespace `OEIS.A355872`. The construction over an arbitrary commutative ring, its compatibility with reduction of coefficients, and the congruence for the whole series modulo 16 are in `lean/Families/BilateralTheta.lean` (namespace `Families.BilateralTheta`). The theorems in the table assume only equation (1), as stated in the entry; no congruence is assumed. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
Equation (1), written coefficient by coefficient, is equivalent to $\Theta(A) = x$	`literal_source_iff`
Theorem 3.1(a) over $\mathbb{Z}$	`source_existence_uniqueness`, `canonical_literal_source`
Theorem 3.1(b)	`canonical_sparse_support`
Theorem 3.1(c), for every integer series satisfying (1)	`source_mod_sixteen`
Corollary 3.2	`source_all_three`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that

statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1). The principal theorems are stated once more, using only Mathlib's definitions, in the file `lean/Challenges/Theta.lean`, and Lean's Comparator tool confirms that the proved theorems have exactly these statements.

6 Discussion

Scope. Theorem 3.1 determines every coefficient modulo 16, so it settles the three conjectures of the entry completely. It says nothing about the size of $a(n)$; the asymptotic formula and the product formulas recorded in the entry are not addressed.

Higher powers of 2. The same mechanism does not immediately give more: modulo 32 the fourth power of P is no longer zero, and more terms of the bilateral sum survive. A computer calculation suggests that the residues remain periodic with period 4: for $n < 200$, $a(n)$ modulo 32 repeats $2, 14, 18, 30$, and modulo 64 it repeats $2, 14, 50, 62$, both from $n = 1$. It would be interesting to describe $a(n)$ modulo every power of 2.

Related equations. The argument of this paper applies to any bilateral equation in which, as here, the term with $j = 0$ is the unknown series and all other terms carry a positive power of x: existence and uniqueness then hold over every ring, and congruences can be found by exhibiting a solution of the reduced equation.

About this paper

How the work was done. This paper is one of 29 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [7]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [7], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and its conjectures are due to Paul D. Hanna [1]. The asymptotic formula mentioned in the introduction was added to the entry by Vaclav Kotesovec.

References

[1] P. D. Hanna, entry A355872 (G.f. $A(x)$ satisfies: $x = \sum_{n\in\mathbb{Z}}(-x)^{n^2}A(x)^{(n-1)^2}$), *The On-Line Encyclopedia of Integer Sequences*, August 2022. `https://oeis.org/A355872`. Text as of October 2026.

[2] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[4] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[5] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[6] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `56a005855ea1`.

[7] D. D. Goyal, *Proofs and corrections of OEIS conjectures*, repository, 2026. `https://github.com/dvgyl/oeis-proofs`.